Bus-Level Load Reconstruction — Validation Results

From ERCOT Coast to Statewide Texas: Methodological Corrections, Model Choice, Feature Engineering, and Validation Against All Eight ERCOT Weather Zones

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Abstract

This report summarizes a series of methodological corrections and validation extensions applied to the bus-level weather-dependent load reconstruction pipeline for the ERCOT Coast zone (200-bus synthetic grid, IECC climate zone 2A), following review comments on the draft paper. Two independent defects were identified and corrected: (1) a temporal train/test leakage in the original random-split evaluation, and (2) a geography bug in the weather station used to train and drive every model. On top of these fixes, a head-to-head comparison of RandomForest, XGBoost, and LightGBM, and a feature-engineering ablation study were carried out, all under a single consistent, leakage-free evaluation protocol. A final validation pass compares every metric across all six pipeline versions produced. The corrected pipeline improves the Pearson correlation against ERCOT Coast actual load from 0.8818 to 0.9022, while providing an honest (non-inflated) picture of per-archetype predictive accuracy — though not every secondary validation metric improves monotonically, a nuance discussed in Section 7.

A second phase of work (Sections 9–12) extends the corrected, feature-engineered LightGBM pipeline from the single Coast case study to all five Texas IECC climate zones (254 counties, on a new, denser synthetic grid), for both the residential and commercial stacks. This extension reuses the Coast methodology as-is and is cross-checked against the already-validated Coast pipeline at $r = 0.9966$ (residential) and $r = 0.9993$ (commercial). A stale RES_SCALE/COM_SCALE calibration was found and fixed while doing this (Section 10, resolving the first open item from the previous phase), and a structural bus-coverage gap in the commercial ZCTA-to-bus spatial join was identified but left unresolved (Section 9). A second, independent, out-of-Coast validation against real ERCOT load was obtained for IECC zone 3B (West Texas/Permian) against ERCOT's own WEST and FAR WEST weather zones ($r = 0.71$, materially weaker than Coast, most likely because West Texas load carries a larger and more weather-independent industrial share that this pipeline does not model at all). Finally, summing all five zones and comparing against ERCOT's true statewide total load yields the strongest system-level correlation obtained in this project so far ($r = 0.9221$ uncalibrated, $r = 0.9231$ with Coast's calibration applied statewide) — a genuinely new, global result, though at that stage still without an independent statewide calibration or any industrial model outside Coast.

A third phase (Sections 13–15) removes the three remaining structural limitations. The winner-takes-all spatial join underlying every earlier number — one census tract or ZIP-code area to one bus — is replaced by an area-split allocation over the full 6,717-bus grid, conserving 100% of Texas housing units while raising the share of load-carrying buses receiving commercial load from 20.1% to 96.1% and lifting bus-level rank agreement with the grid case's own per-bus load from $\rho = 0.133$ to 0.475 — an improvement invisible to every zone-aggregate metric, which is itself an argument for reporting a bus-level statistic. Validation is then extended to *all eight* ERCOT weather zones via a per-bus assignment carried by the grid model itself, rather than the GIS shapefiles previously assumed necessary; the zone-3B finding is sharpened, locating the weakness specifically in West Texas ($r = 0.629$ for FAR WEST) rather than in dry climates generally. Calibration is finally refitted per zone with a *data-driven* industrial constant, scored

out-of-sample: Coast's factors do not transfer (applied statewide with no industrial term they already reach 105% of ERCOT's true peak), the refitted residential factors collapse from 1.6–1.9 to 1.07–1.36, and the missing weather-insensitive load is quantified at 18,079 MW statewide, or 24.7% of system peak.

Contents

1 Context **6**

2 Baseline: the Original Pipeline **6**

3 Temporal Train/Test Leakage **6**
3.1 Problem . . . 6
3.2 A Naive Fix That Did Not Work . . . 7
3.3 Fix: Blocked Seasonal 4-Fold Cross-Validation . . . 7
3.4 Results . . . 7

4 Weather Station Geography Correction **8**
4.1 Bug Discovery . . . 8
4.2 Fix . . . 8
4.3 Results . . . 9
4.4 A Related Finding: the True Coast-Zone Footprint . . . 9

5 Model Comparison: RandomForest vs. XGBoost vs. LightGBM (Point 6) **10**
5.1 Setup . . . 10
5.2 Per-Archetype Accuracy and Training Time . . . 10
5.3 System-Level Check . . . 11
5.4 Why No LSTM Was Implemented in Code . . . 12

6 Feature Engineering **12**
6.1 Ablation Design . . . 12
6.2 Results . . . 13
6.3 System-Level Check . . . 13

7 Validation Metrics Beyond Aggregate Correlation (Point 5): Full Comparison **14**
7.1 Pearson Correlation, Peak, and Load Factor . . . 14
7.2 Coincidence and Diversity Factor . . . 15
7.3 Load Duration Curve . . . 15
7.4 Diurnal Shape . . . 16
7.5 Seasonal Monthly Bias . . . 17
7.6 Spatial Consistency Proxy . . . 17
7.7 Out of Scope: Comparison Against Prior Synthetic-Grid Methods . . . 17
7.8 Section Takeaway . . . 17

8 Combined Progression of Results **18**

9 Statewide Extension to All Five Texas IECC Climate Zones **18**
9.1 Zone Definitions and New Bus Grid . . . 18
9.2 Two New Geography Issues Found . . . 19
9.3 Methodology . . . 19
9.4 Honest Accuracy Results . . . 20
9.5 Cross-Check Against the Already-Validated Coast Pipeline . . . 21
9.6 Statewide Totals (Uncalibrated) . . . 21

10 Coast Recalibration **21**
10.1 Bug Discovery . . . 21
10.2 Fix and Results . . . 22

11 Validation Beyond Coast: an ERCOT Weather-Zone Crosswalk **22**
11.1 Zone 3B: West Texas / Permian Basin . . . 23

12 A Global, Statewide Validation Result **24**

13 Spatial Allocation Refinement: From Nearest-Bus to Area-Split **25**
13.1 Method . . . 25
13.2 Results . . . 26
13.3 A Negative Result Worth Recording . . . 27

14 ERCOT Weather-Zone Validation: All Eight Zones **27**
14.1 Validating the Crosswalk Before Using It . . . 27
14.2 Results . . . 28

15 Per-Zone Calibration with a Fitted Industrial Constant **28**
15.1 Method . . . 28
15.2 Results . . . 29
15.3 Two Caveats, One Unresolved . . . 31

16 Open Items and Next Steps **31**
16.1 What Would Help This Move Faster . . . 32

17 Conclusion **33**

List of Figures

1 R^2 per archetype: original random split (leaky) versus seasonal blocked CV mean (honest), still using the (subsequently found to be incorrect) original weather station. 8
2 Seasonal-CV mean R^2 per archetype: wrong weather (Upton County) vs. corrected weather (Coast-zone population-weighted blend). . . . 9
3 Seasonal-CV R^2 per archetype: RandomForest vs. XGBoost vs. LightGBM (corrected Coast weather). . . . 10
4 Mean feature importance (normalized to sum to 1 per model) across all 20 archetypes: RandomForest vs. XGBoost vs. LightGBM. . . . 11
5 System-level Pearson correlation against ERCOT Coast, across every fix applied so far (leakage, weather geography, model choice). . . . 11

6 Ablation: mean seasonal-CV R^2 by sector, incremental feature additions (LightGBM, corrected Coast weather). . . . 13
7 Load duration curve across pipeline versions. The systematic vertical gap between every estimated curve and ERCOT actual reflects the not-yet-re-optimized calibration scale factors, not a shape mismatch. . . . 15
8 Average diurnal load shape by weekday/weekend × summer/winter, ERCOT actual vs. Original vs. Weather-fixed vs. Feature-engineered. . . . 16
9 Monthly bias (%) vs. ERCOT actual, across pipeline versions. . . . 17
10 Mean honest seasonal-CV R^2 by IECC zone, residential vs. commercial. . . . 20
11 Old stale calibration vs. newly recalibrated total, both applied to the current best (`_featureeng`) pipeline, vs. ERCOT Coast actual. . . . 22
12 Zone 3B reconstruction (in-ERCOT county subset, uncalibrated) vs. ERCOT WEST+FAR WEST actual load, 2018. . . . 23
13 Statewide reconstruction (five IECC zones, residential + commercial) vs. ERCOT's actual statewide total load, 2018. . . . 24
14 Left: share of load-carrying buses receiving commercial floor space, by ERCOT weather zone. Right: bus-level rank agreement with the case's own `Load_MW`. . . . 26
15 Left: the fitted weather-insensitive baseload as a share of each zone's ERCOT peak. Right: out-of-sample error with and without the industrial term. . . . 29
16 Calibrated bus-level reconstruction aggregated to each ERCOT weather zone, 2018. Daily means are plotted for legibility; the reported r is computed on hourly data. . . . 30

List of Tables

1 Mean seasonal-CV R^2 versus the original random-split R^2, wrong (Upton County) weather station. . . . 7
2 Population (housing-unit) weights for the corrected Coast-zone weather blend. . . . 9
3 Mean seasonal-CV R^2, wrong vs. corrected weather. . . . 9
4 Mean seasonal-CV R^2 and mean per-archetype/fold fit time, by model. . . . 10
5 Mean seasonal-CV R^2 by sector, per feature set. . . . 13
6 System-level validation, baseline vs. feature-engineered LightGBM. . . . 14
7 Pearson correlation, peak, and load factor across all six pipeline versions. . . . 14
8 Coincidence factor, residential-only, commercial-only, and combined bus sets, all six versions. . . . 15
9 Correlation of average diurnal shape (not level) against ERCOT actual, all six versions. 16
10 Full progression across every correction applied. Pearson is measured against ERCOT Coast actual load, whose 2018 peak is 20,270 MW. *leakage-inflated, not comparable to the honest values below them. . . . 18
11 The five Texas IECC climate zones covering all 254 counties. The bus counts are those of the initial nearest-bus mapping; the authoritative bus table adopted in Section 13 covers the full grid of 6,717 buses (2A 3,580, 2B 246, 3A 2,142, 3B 625, 4B 124), with identical county assignments. . . . 19

12 Honest seasonal-CV mean R^2, ranked by residential difficulty. Zone 2A (Coast's own climate) is clearly easiest; drier/more continental zones are harder, plausibly a mix of a more complex weather-load relationship (real heating seasons, not purely cooling-dominated) and smaller sample sizes. Values are from the current run; they shifted by up to 0.014 from the first statewide pass because the area-split allocation (Section 13) slightly changes each zone's county weather weights. Model choice, features and protocol are unchanged. 20
13 Cross-check: the new statewide methodology, restricted to the original 12 Coast counties, reproduces the already-validated Coast-only reconstruction closely on both stacks — the shape correlation is what this test is for, and it is essentially perfect. The level gap is larger on residential (+14.6%) than commercial (+2.8%) because the two footprints are not the same set: the full grid places 1,505 buses in those 12 counties against the old grid's 200, so the subset captures more of the counties' housing. The zone-wide weather blend (98 counties for the 2A run) diluting the Coast-specific 12-county blend also contributes. 21
14 Per-zone and statewide peak load, residential and commercial, uncalibrated (no RES_SCALE/COM_SCALE applied) and with no industrial component anywhere except Coast's flat placeholder (not included in this table). The statewide row is the peak of the summed statewide series, not the sum of the per-zone peaks, which is why it falls below the column total: the five zones do not peak in the same hour. . . 21
15 ERCOT Coast actual peak: 20,269.8 MW. Pearson correlation is near scale-invariant and barely moves, but the old calibration was overcorrecting for a weaker, pre-fix pipeline: the absolute error nearly halves under the new calibration. **RES_SCALE=1.6, COM_SCALE=1.3 is the number to cite going forward.** 22
16 Zone 3B vs. Coast for reference: Coast's uncalibrated peak coverage is ∼57% and its Pearson correlation is ∼0.90. The ∼57% figure is for the original 200-bus Coast-only pipeline; the same quantity measured on the statewide 6,717-bus grid is 65.3% (Table 20), the difference coming from the area-split allocation and the larger bus set, not from a change in the load models. 24
17 The statewide aggregate correlates *more strongly* with ERCOT's true system total than any single zone does individually — including Coast. 25
18 Effect of the area-split allocation, scored against the Texas 7k case's own per-bus `Load_MW`. Spearman (rank) rather than Pearson is the meaningful statistic here: `Load_MW` also contains industrial demand, so a handful of very large industrial buses dominate any linear correlation. 26
19 The synthetic case's EAST and NORTH area boundaries do not coincide with ERCOT's. Those two zones are reported throughout with a boundary-mismatch flag: their shape statistics remain interpretable, their magnitude statistics do not. 27
20 Uncalibrated reconstruction (no scale factors, no industrial model) against ERCOT actual load, 2018. † boundary mismatch: peak coverage is not meaningful for these two zones. 28
21 Per-zone calibration. "from" gives the blocked-seasonal-CV MAE of the Coast-style two-parameter fit with no industrial term, for comparison. † boundary mismatch, reported for completeness. 29

1 Context

The goal of this project is to reproduce and then improve on the bus-level, weather-dependent load reconstruction framework described in Kang, Baembitov, Safdarian, Snodgrass, and Overbye, applied to the ERCOT Coast weather zone (a 200-bus synthetic grid, IECC climate zone 2A, covering the Houston/Galveston area). The pipeline combines NREL ResStock/ComStock building archetype simulations, Census and County Business Patterns (CBP) data, and weather-driven machine learning models to reconstruct hourly, bus-level electricity load, which is then validated against ERCOT's published actual load for the Coast zone.

The review of the draft paper raised two points that this report addresses directly:

- **Point 5**: the existing validation is too system-level (a single aggregate Pearson correlation), and should be supplemented with additional validation angles (coincidence factor, diversity factor, load duration curves, seasonal behavior, diurnal shape).
- **Point 6**: reviewers will ask why RandomForest was used instead of XGBoost, LightGBM, or an LSTM, and a justification or comparison is needed.

While investigating these two points, two additional, independent defects were found in the original pipeline and corrected; both are documented here because they materially changed the results. Section 7 then revisits point 5's validation metrics one final time, across every pipeline version produced in this report, rather than just the first two.

2 Baseline: the Original Pipeline

The original pipeline (notebooks `residential.ipynb`, `commercial.ipynb`, `plot_all.ipynb`) reproduces the published paper's results exactly: 5 RandomForest models for residential archetypes and 15 for commercial archetypes, trained with an ordinary random 80/20 train/test split, feeding a bus-level reconstruction that is compared to ERCOT's Coast-zone actual load after applying a flat industrial baseline (4500 MW) and grid-searched residential/commercial scale factors (`RES_SCALE` = 1.9, `COM_SCALE` = 1.3). This baseline reaches a Pearson correlation of **0.8818** against ERCOT Coast, with a reported peak of 24,418 MW versus ERCOT's actual peak of 20,270 MW.

Two limitations of this baseline motivated the work below: the R^2 scores reported for the underlying models (residential archetypes 0.85–0.95, commercial archetypes 0.93–0.98) were measured with a random train/test split on strongly autocorrelated hourly data, and the single weather station used to drive every model turned out — on investigation — not to be representative of the Coast zone at all (Section 4).

3 Temporal Train/Test Leakage

3.1 Problem

Random shuffling of hourly weather/load pairs before splitting into train and test sets allows the model to see near-duplicates of test points during training: adjacent hours have almost identical weather and almost identical load, so a randomly assigned test hour is frequently sitting right next to a training hour with essentially the same answer. This inflates the reported R^2 without reflecting real predictive skill on genuinely unseen conditions.

3.2 A Naive Fix That Did Not Work

A single chronological 80/20 holdout (train on January–October, test on the remaining October–December) was tried first. This produced *negative* R^2 for several archetypes. The root cause: the test tail contains November and December, months entirely absent from the training data, and RandomForest — a tree-based, non-extrapolating model — cannot generalize past the range of feature values it was trained on. Cyclically encoding the calendar features (sine/cosine of month, hour, day-of-week instead of raw ordinal values) was tried as a fix and did not reliably help: it improved some archetypes and worsened others, because the deeper issue is that a single calendar year only contains one November, not an encoding problem.

3.3 Fix: Blocked Seasonal 4-Fold Cross-Validation

The evaluation protocol adopted instead trains on three seasons and tests on the fourth held-out season, rotating through all four (winter/spring/summer/fall). Every fold has enough seasonal diversity to learn from, and only the two seasonal boundaries per fold introduce any adjacency between train and test, versus every single hour under full random shuffling. Final production models are then trained on the full year, since the cross-validation already estimated their honest generalization error — standard practice.

3.4 Results

	Residential (mean of 5)	Commercial (mean of 15)
Original (random split)	0.90	0.97
Honest (seasonal blocked CV)	**0.41**	**0.79**

Table 1: Mean seasonal-CV R^2 versus the original random-split R^2, wrong (Upton County) weather station.

Despite the large drop in per-archetype R^2, the system-level Pearson correlation against ERCOT Coast barely moved: **0.8818** $\rightarrow$ **0.8907**. Aggregating thousands of individual building predictions cancels idiosyncratic per-building error while the shared weather-driven signal reinforces itself — the leak inflated a secondary metric (per-archetype precision), not the paper's headline system-level claim.

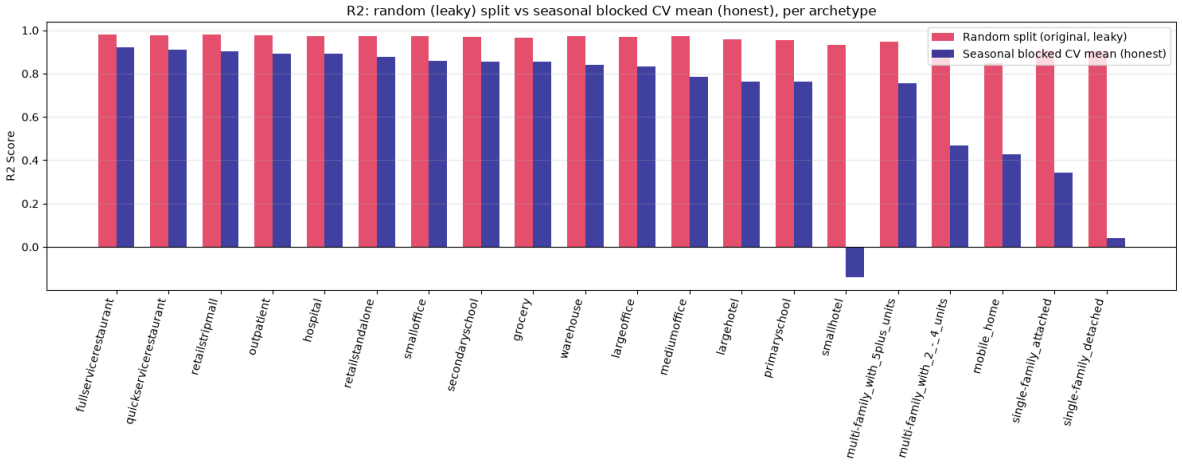


Figure 1: R^2 per archetype: original random split (leaky) versus seasonal blocked CV mean (honest), still using the (subsequently found to be incorrect) original weather station.

4 Weather Station Geography Correction

4.1 Bug Discovery

While investigating the residential weakness identified in Section 3, the single weather station used throughout the pipeline (station ID `G4804610`) was found not to belong to the Coast zone at all. NREL indexes its weather files by *county*, not by PUMA code, but the original notebooks build weather station IDs from PUMA codes. Of roughly 50 constructed IDs, only two ever resolved on NREL's S3 bucket — by pure numeric coincidence with two unrelated counties' FIPS codes. Decoded via the official Census county reference, `G4804610` corresponds to **Upton County**, in the Permian Basin of West Texas, approximately 800 km from Houston, and `G4804630` to **Uvalde County** in South Texas. Neither is one of the twelve real Coast-zone counties.

This was confirmed directly in the weather data: Upton County averages 50.6% relative humidity and spans −11.7 to 42.2 °C, versus Harris County (Houston)'s 76.4% average relative humidity and −7.2 to 37.8 °C range — a meaningfully drier, more continental climate than the humid Gulf Coast the pipeline is meant to represent. Every model in the pipeline had been trained and driven by the wrong region's weather.

4.2 Fix

NREL does provide a correct weather file for every Texas county. A population-weighted blend across the twelve real Coast-zone counties was built, weighted by residential housing units (already-cached Census B25024 data, no new API calls needed):

County	Weight	County	Weight
Harris	66.84%	Victoria	1.40%
Fort Bend	9.51%	Matagorda	0.74%
Montgomery	8.20%	Wharton	0.67%
Galveston	5.54%	Waller	0.64%
Brazoria	5.15%	Chambers	0.60%
		Calhoun	0.45%
		Jackson	0.25%

Table 2: Population (housing-unit) weights for the corrected Coast-zone weather blend.

All 20 archetype models were retrained through the identical honest (seasonal blocked CV, full-year final training) protocol from Section 3.

4.3 Results

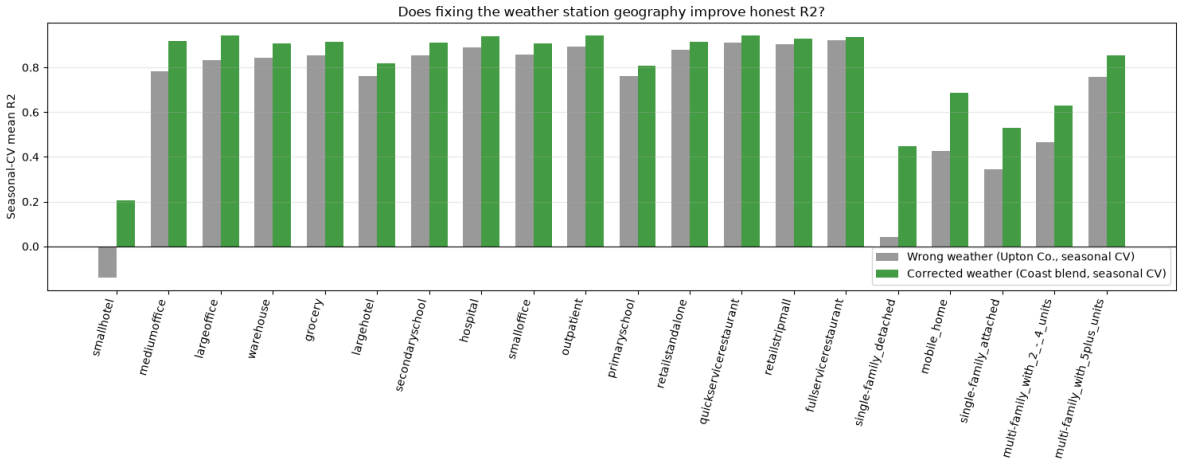


Figure 2: Seasonal-CV mean R^2 per archetype: wrong weather (Upton County) vs. corrected weather (Coast-zone population-weighted blend).

	Residential	Commercial
Wrong weather (honest split)	0.41	0.79
Corrected weather (honest split)	**0.63**	**0.86**

Table 3: Mean seasonal-CV R^2, wrong vs. corrected weather.

The residential gain (+0.22) is large, confirming that a substantial part of the earlier winter/spring weakness was a geography artifact. The system-level Pearson correlation against ERCOT Coast improved again: **0.8907** $\rightarrow$ **0.8986**, with a reported peak of 26,115 MW.

4.4 A Related Finding: the True Coast-Zone Footprint

Cross-referencing the original 56-PUMA Coast zone selection against the official Census 2010 Tract-to-PUMA relationship file shows that the full PUMA-based zone definition actually spans **21 coun-**

ties, while the bus-assignment file (`tract_to_bus_mapping.csv`) used both for building counts and for the weather correction above only covers **12** of those 21. The nine missing counties (Austin, Colorado, DeWitt, Goliad, Gonzales, Karnes, Lavaca, Liberty, Wilson) are peripheral, inland counties without any bus of the 200-bus grid nearby. The 12-county housing-unit total (2,645,171) matches the paper's already-validated population figure (2,641,624 housing units) closely, confirming the 12-county set is the correct, intentional footprint of the actual synthetic grid — not an error in the weather correction, but worth stating explicitly in the paper that the model's "Coast zone" does not claim to cover ERCOT's full official Coast weather-zone boundary, only the actual footprint of the 200-bus grid.

5 Model Comparison: RandomForest vs. XGBoost vs. LightGBM (Point 6)

5.1 Setup

All three models were run through the identical seasonal blocked 4-fold CV protocol, on the corrected Coast-zone weather, with reasonable out-of-the-box hyperparameters (no tuning search) — a fair baseline-vs-baseline comparison, not a claim of best-possible performance for any model.

5.2 Per-Archetype Accuracy and Training Time

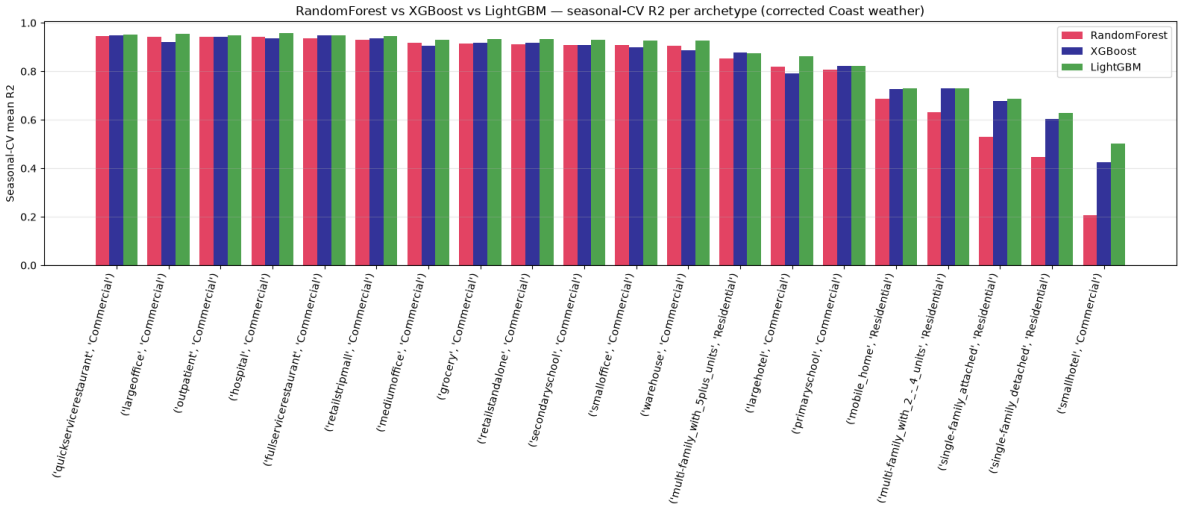


Figure 3: Seasonal-CV R^2 per archetype: RandomForest vs. XGBoost vs. LightGBM (corrected Coast weather).

	Residential R^2	Commercial R^2	Mean fit time (s)
RandomForest	0.629	0.863	0.329
XGBoost	0.724	0.874	**0.088**
LightGBM	**0.730**	**0.899**	0.372

Table 4: Mean seasonal-CV R^2 and mean per-archetype/fold fit time, by model.

LightGBM wins on accuracy in both sectors; XGBoost trains roughly 4× faster than the other two. The residential gain from switching model alone (0.629 → 0.730) is comparable in magnitude to the earlier weather-station fix's gain — a second, independent lever on the residential weakness.

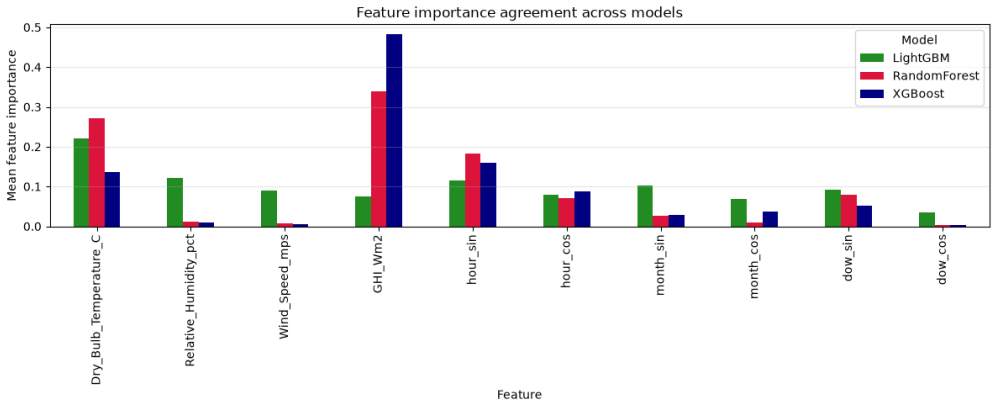


Figure 4: Mean feature importance (normalized to sum to 1 per model) across all 20 archetypes: RandomForest vs. XGBoost vs. LightGBM.

5.3 System-Level Check

Final full-year production models were trained for all three algorithms, bus-level profiles regenerated, and the ERCOT Coast Pearson correlation re-checked — the same "does it survive aggregation" test applied to the weather fix.

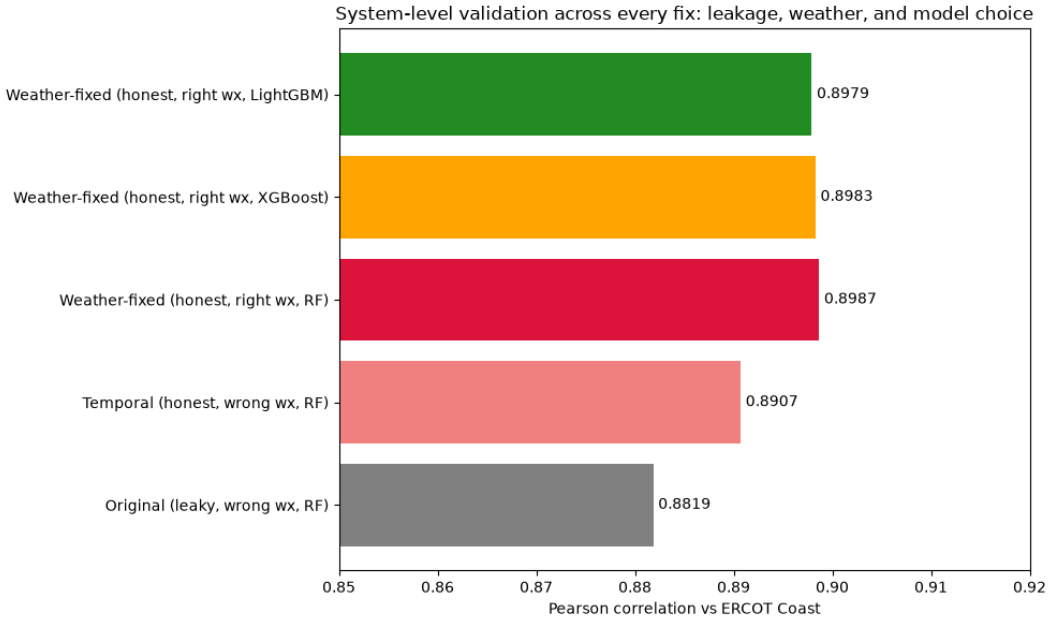


Figure 5: System-level Pearson correlation against ERCOT Coast, across every fix applied so far (leakage, weather geography, model choice).

Despite LightGBM's substantially better per-archetype R^2, all three models land within 0.0008 of each other at the system level (RF 0.8986, XGBoost 0.8983, LightGBM 0.8979) — statistically

indistinguishable. **Model choice affects per-building precision, not the aggregate system-level validation number.** This aggregation-washes-out-precision pattern mirrors the finding for temporal leakage in Section 3.

This result supports two equally defensible framings for the paper: (i) retain RandomForest, since model choice does not affect the headline validation and RF is the simpler, more interpretable choice, or (ii) adopt LightGBM, since it provides a materially better per-archetype fit at zero system-level cost. Both are stated here; the final choice of framing is left open (Section 16).

5.4 Why No LSTM Was Implemented in Code

A recurrent/sequence model such as an LSTM learns from a window of past timesteps, which would mainly help if the target depends on its own recent history beyond what the instantaneous input features encode — for example, thermal inertia in a building (a preceding heat wave keeping indoor temperature elevated after outdoor temperature drops). This is a plausible mechanism, not implemented in code this round because (i) it requires a different data representation (contiguous sequence windows, which interacts awkwardly with the blocked seasonal CV), (ii) it requires a new, heavy dependency (PyTorch) and materially longer training across 20 archetypes and 4 folds, and (iii) a cheaper, incremental test of the same hypothesis is available: adding lag and rolling-window weather features to the existing tabular models. That cheaper test was run next, in Section 6, and confirms the hypothesis.

6 Feature Engineering

6.1 Ablation Design

Using LightGBM (the Section 5 winner) held fixed, feature groups were added incrementally to isolate which additions actually help, under the same seasonal blocked CV:

- **A (baseline)**: temperature, relative humidity, wind speed, GHI + cyclical hour/month/day-of-week (10 features).
- **B = A + free unused weather columns**: wind direction (cyclically encoded), direct normal radiation (DNI), diffuse horizontal radiation (DHI) — already present in the cached weather files, unused until now.
- **C = B + degree-hours**: heating and cooling degree-hours at an 18 °C balance point — the traditional load-forecasting feature.
- **D = C + lag/rolling temperature**: temperature 1 h and 3 h prior, and a 24 h rolling mean — a direct, cheap test of the thermal-inertia hypothesis from Section 5.
- **E = D + US federal holiday indicator (2018)**.

6.2 Results

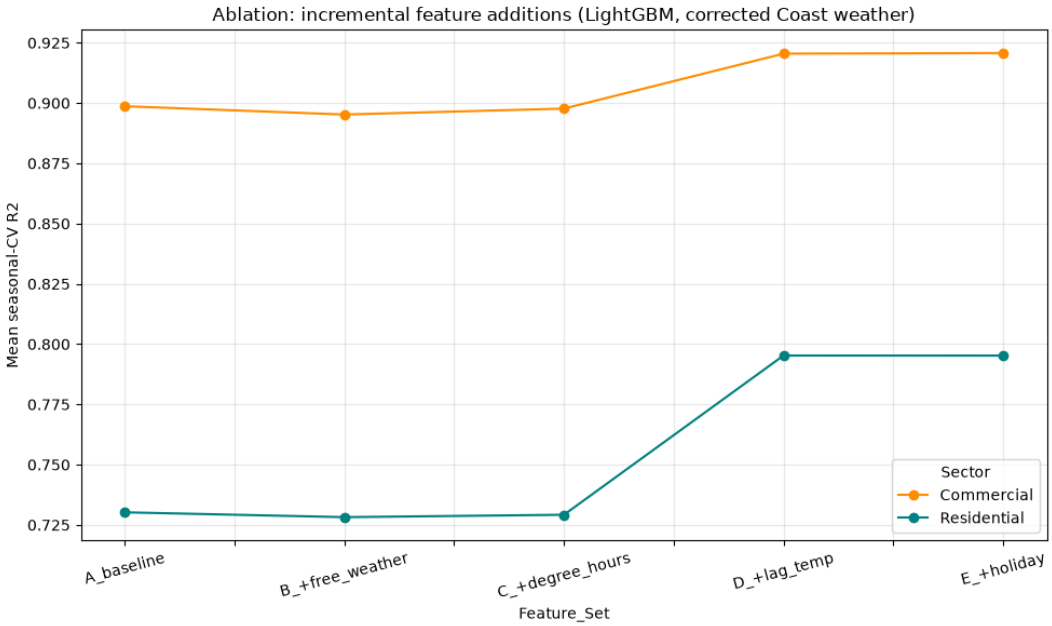


Figure 6: Ablation: mean seasonal-CV R^2 by sector, incremental feature additions (LightGBM, corrected Coast weather).

Feature set	Residential R^2	Commercial R^2
A: baseline	0.730	0.899
B: + free weather columns	0.728	0.895
C: + degree-hours	0.729	0.898
D: + lag/rolling temperature	**0.795**	**0.921**
E: + holiday indicator	0.795	0.921

Table 5: Mean seasonal-CV R^2 by sector, per feature set.

The free unused weather columns and degree-hours contribute essentially nothing. Lagged/rolling temperature produces the single largest gain seen in any experiment in this report (residential 0.730 $\rightarrow$ 0.795), confirming the thermal-inertia hypothesis directly and cheaply, without a new model family or dependency. The holiday indicator adds nothing further on top.

6.3 System-Level Check

Final full-year LightGBM models were trained with feature set E, bus-level profiles regenerated, and the ERCOT correlation re-checked:

	Pearson vs. ERCOT Coast	Peak (MW)
LightGBM, baseline features (A)	0.8979	25,735
LightGBM, feature set E	**0.9022**	25,578

Table 6: System-level validation, baseline vs. feature-engineered LightGBM.

Unlike the model-choice comparison in Section 5, this change *does* move the system-level number (+0.0043) — evidence that the lag/thermal-inertia signal is a genuine physical effect that partially survives aggregation, not merely per-archetype noise reduction.

7 Validation Metrics Beyond Aggregate Correlation (Point 5): Full Comparison

Point 5 of the review flagged the existing validation as too system-level. This section revisits every metric proposed there — coincidence/diversity factor, load duration curve, diurnal shape, seasonal bias, and a spatial-consistency proxy — but now computed across **all six pipeline versions** produced in this report: Original, Temporal, Weather-fixed, XGBoost, LightGBM, and Feature-engineered. All six use the same, not-yet-re-optimized calibration (`RES_SCALE` = 1.9, `COM_SCALE` = 1.3, `IND_BASE` = 4500 MW — see Section 16). Figures are limited to ERCOT actual + Original + Weather-fixed + Feature-engineered for legibility, since Section 5 already established that model choice alone does not visibly change the aggregate shape; full numeric tables include all six versions.

This section's headline finding is that the story is not uniformly monotonic. The Pearson correlation improves with every fix, but several secondary metrics do not — a more nuanced and more defensible picture than "every fix improved everything."

7.1 Pearson Correlation, Peak, and Load Factor

Pipeline	Pearson vs. ERCOT	Peak (MW)	Load Factor
Original	0.8818	24,418	0.582
Temporal	0.8907	24,604	0.578
Weather-fixed	0.8986	26,115	0.544
XGBoost	0.8983	26,279	0.541
LightGBM	0.8979	25,735	0.552
Feature-engineered	**0.9022**	25,578	0.556
ERCOT Actual	—	20,270	0.600

Table 7: Pearson correlation, peak, and load factor across all six pipeline versions.

Pearson correlation improves essentially monotonically. Load factor does **not**: Original (0.582) is actually the closest of all six to ERCOT's 0.600, and the weather fix moves it further away (0.544) before feature engineering partially recovers it (0.556). The corrections that improved hour-by-hour correlation made the estimated stack slightly peakier relative to ERCOT actual than the uncorrected pipeline was.

7.2 Coincidence and Diversity Factor

Pipeline	CF, Residential	CF, Commercial	CF, Combined
Original	0.9983	0.9984	0.9625
Temporal	0.9957	0.9967	0.9789
Weather-fixed	0.9996	0.9961	0.9857
XGBoost	1.0000	0.9985	0.9896
LightGBM	0.9997	0.9981	0.9885
Feature-engineered	1.0000	0.9985	0.9843

Table 8: Coincidence factor, residential-only, commercial-only, and combined bus sets, all six versions.

Within-sector CF stays pinned near 1.0 across every version — a structural property of the current single, zone-wide weather series design (Section 16), unaffected by any split, weather, model, or feature fix tried so far. The *combined* (residential + commercial) CF trends from 0.963 toward 0.99 as fixes are applied, meaning *less* diversity between residential and commercial peak timing, not more — plausibly because better models converge more tightly on the same dominant driver (temperature), so the two sectors' peaks align more closely.

7.3 Load Duration Curve

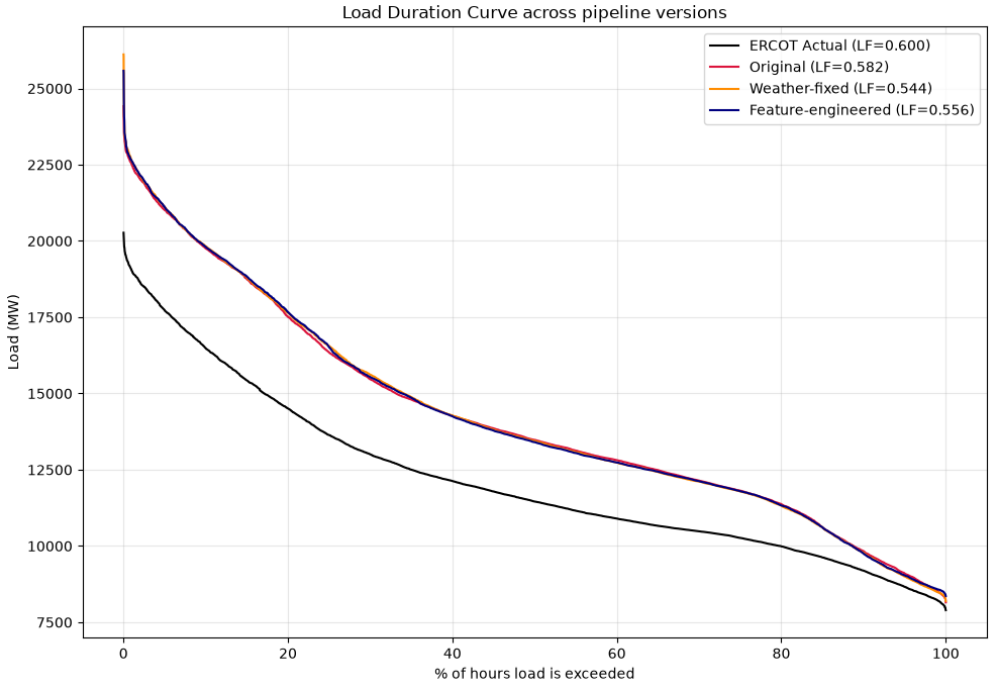


Figure 7: Load duration curve across pipeline versions. The systematic vertical gap between every estimated curve and ERCOT actual reflects the not-yet-re-optimized calibration scale factors, not a shape mismatch.

7.4 Diurnal Shape

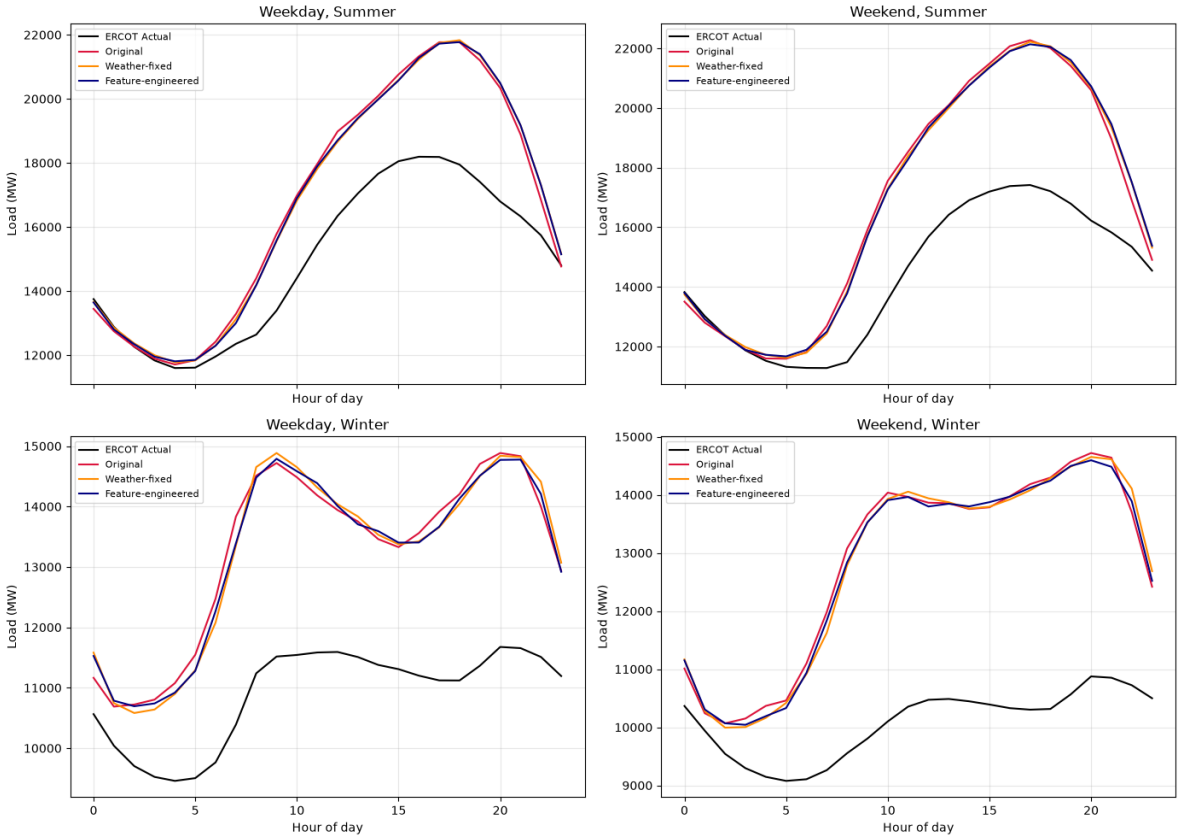


Figure 8: Average diurnal load shape by weekday/weekend × summer/winter, ERCOT actual vs. Original vs. Weather-fixed vs. Feature-engineered.

Pipeline	Wd. Summer	We. Summer	Wd. Winter	We. Winter
Original	0.9757	0.9483	0.8739	0.7668
Temporal	0.9808	0.9610	0.9192	0.8047
Weather-fixed	0.9807	0.9600	**0.9223**	**0.8116**
XGBoost	0.9809	0.9600	0.9173	0.8090
LightGBM	0.9808	0.9614	0.9192	0.7997
Feature-engineered	0.9801	0.9610	0.9151	0.7978

Table 9: Correlation of average diurnal shape (not level) against ERCOT actual, all six versions.

The winter-weekend condition — the single weakest metric identified anywhere in this report — improves substantially from Original (0.767) to a peak at Weather-fixed (0.812), but then **drifts back down** through XGBoost, LightGBM, and Feature-engineered (0.798), still comfortably above the original value but below the weather-fix peak. The temporal and weather-geography fixes are what actually repaired this particular metric; the later model-choice and feature-engineering refinements improved per-archetype accuracy and the overall Pearson correlation without further improving this specific shape sub-metric — in fact trading a small amount of it away.

7.5 Seasonal Monthly Bias

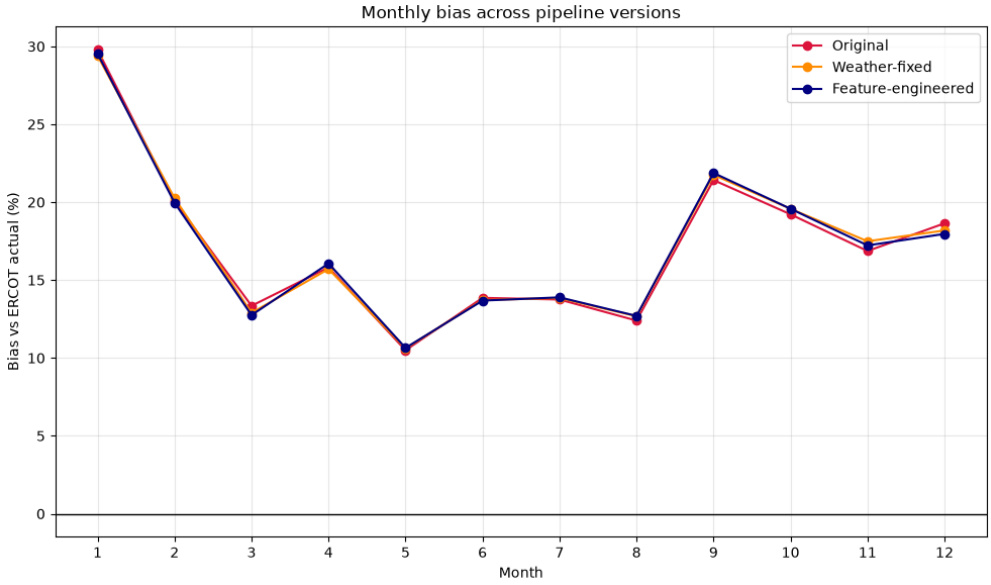


Figure 9: Monthly bias (%) vs. ERCOT actual, across pipeline versions.

Bias is positive in every month for every version (range: +10.5% in May to +29.8% in January), and the three plotted versions track each other closely — differences between versions are mostly under one percentage point per month. This is consistent with a calibration-scale issue rather than a shape issue: the pipeline over-predicts uniformly across the year, which the RES_SCALE/COM_SCALE recalibration (Section 16) is expected to address directly.

7.6 Spatial Consistency Proxy

Correlation between per-bus annual energy and per-bus building/sqft count is **effectively invariant** across all six pipeline versions: residential is identical to four decimal places (0.9479), commercial varies by one unit in the fourth decimal (0.8917–0.8918). This confirms the proxy is insensitive to every fix and refinement tried — expected, since building counts and the sqft mapping are unchanged Census/ArcGIS inputs across all six versions, and this metric was always primarily a plumbing sanity check rather than an independent spatial validation.

7.7 Out of Scope: Comparison Against Prior Synthetic-Grid Methods

Review point 9 requested a quantitative comparison against a prior proportional load allocation method (referenced as "Hanyue's method"). This has not been implemented — the exact method description is not yet available and is needed before this comparison can be built.

7.8 Section Takeaway

The temporal-leakage and weather-geography fixes (Sections 3 and 4) are the corrections that improved the secondary shape metrics (load duration curve shape, winter-weekend diurnal match) as well as the headline correlation. The later model-choice and feature-engineering refinements (Sections 5 and 6) mainly improved per-archetype accuracy and the aggregate Pearson number,

while leaving flat — or in one case (winter-weekend diurnal shape) slightly trading off — some of these secondary metrics. Reporting this honestly, rather than claiming uniform improvement, is a stronger and more defensible position for reviewers.

8 Combined Progression of Results

Pipeline version	Res. R^2	Com. R^2	Pearson	Peak (MW)
Original (leaky split, wrong weather, RF)	0.90*	0.97*	0.8818	24,418
+ honest temporal split	0.41	0.79	0.8907	24,604
+ corrected weather geography	0.63	0.86	0.8986	26,115
+ LightGBM (vs. RandomForest)	0.73	0.90	0.8979	25,735
+ feature engineering (lag temperature)	**0.80**	**0.92**	**0.9022**	25,578

Table 10: Full progression across every correction applied. Pearson is measured against ERCOT Coast actual load, whose 2018 peak is 20,270 MW. *leakage-inflated, not comparable to the honest values below them.

Two independent bugs (temporal leakage, weather geography) and two optional improvements (model choice, feature engineering) were tested under one consistent, non-leaking evaluation protocol throughout. The Pearson correlation against ERCOT Coast actual load improved overall from 0.8818 to 0.9022 across the full sequence of changes — though not strictly monotonically: the model-choice step dips slightly (0.8986 → 0.8979) before feature engineering recovers and exceeds it. The per-archetype R^2 figures reported are now honest, non-inflated estimates rather than the originally-reported, leakage-inflated values. As detailed in Section 7, even this overall improvement holds for the headline correlation specifically, not for every secondary validation metric.

9 Statewide Extension to All Five Texas IECC Climate Zones

The previous sections all concern the single, original 200-bus Coast case study (IECC zone 2A). Section 16 of the prior phase of this work listed "extension to all of Texas" as blocked, specifically on a missing bus-to-load spatial mapping (substation locations/topology) for any region beyond the existing 200-bus Coast grid. Two new GIS join files have since been obtained, built for a genuinely new, denser statewide synthetic grid: `tract_to_bus_mapping_statewide_ExportTable.csv` (6,896 census tracts → 3,426 buses, residential) and `zcta_bus_mapping_tx_final.csv` (2,031 ZCTAs → bus, commercial). This is *not* an extension of the original 200-bus Coast grid: a 1 km nearest-substation match against the old Coast file only recovers 72 of the 200 old buses, confirming it is a different, denser grid built independently.

9.1 Zone Definitions and New Bus Grid

An authoritative county → IECC climate-zone crosswalk was built from NREL's own `spatial_tract_lookup_table.csv` (the same geographic reference ResStock itself ships with), rather than a third-party map — deliberately avoiding the exact kind of geography mismatch that caused the original PUMA-vs-county weather bug (Section 4). No Texas county is split across two IECC zones.

Zone	Counties	Buses	Description
2A	79	1,883	Very Hot-Humid (includes the original 12-county Coast footprint)
2B	14	81	Hot-Dry
3A	63	1,219	Warm-Humid (DFW / North / East Texas)
3B	70	194	Warm-Dry (West Texas / Permian Basin)
4B	28	52	Mixed-Dry (Texas Panhandle / Far West)

Table 11: The five Texas IECC climate zones covering all 254 counties. The bus counts are those of the initial nearest-bus mapping; the authoritative bus table adopted in Section 13 covers the full grid of 6,717 buses (2A 3,580, 2B 246, 3A 2,142, 3B 625, 4B 124), with identical county assignments.

9.2 Two New Geography Issues Found

Census tract vintage mismatch. While building the first zone (2A), 740 of its 1,883 buses (39%) initially came out with zero residential building count. Root cause: the new statewide bus mapping is built on *2020* Census tract boundaries (Harris County = 1,115 tracts), while the cached Census B25024 file used throughout Sections 2–6 is on *2010* tract boundaries (Harris County = 786 tracts — correct for, and only for, the original Coast-only file). Fixed by re-fetching Census B25024 on the 2022 ACS5 vintage (confirmed to return exactly 1,115 Harris tracts). After the fix, zero-count buses dropped to under 1% in every zone. This is the same class of bug as the original weather-station geography issue (Section 4) — a silent geography/vintage mismatch — caught the same way, by checking a suspicious diagnostic number instead of trusting a silent join.

Structural commercial bus-coverage gap. Unlike the tract-to-bus join (6,896 tracts for 3,426 buses, ratio > 1, so nearly every bus receives at least one tract), the ZCTA-to-bus join has only 2,031 ZCTAs for 3,426 buses (ratio < 1). Since each ZCTA is matched to its single nearest bus, at most 2,031 buses can ever receive commercial load this way, and in practice only 1,598 (47%) do — compared to 92.5% coverage on the original, much smaller and denser 200-bus Coast grid. This is not a code bug but a structural property of the provided file: it concentrates commercial load onto fewer buses than physically realistic rather than losing or double-counting it (region-level totals are unaffected, confirmed by the cross-check in Section 9.5), but it does mean per-bus commercial values are not reliable for any downstream use that inspects individual buses (e.g. power-flow studies) rather than regional aggregates. *Resolved in Section 13*: an area-split join (100 equal-area parts per ZCTA) raises coverage of load-carrying buses to 96.1%, and confirms the claim made here that region-level totals were unaffected — every zone-aggregate number in this section is unchanged by the fix.

9.3 Methodology

The already-validated Coast methodology (Sections 3–6) is reused as-is, generalized per zone, with no re-derivation of settled conclusions:

- Population-weighted county weather blend (Section 4's fix), generalized from Coast's 12 counties to each zone's full county list.
- Cyclical calendar features plus lagged/rolling temperature (Section 6's winning feature set).
- LightGBM (Section 5's winning model).
- Blocked seasonal 4-fold cross-validation for the honest R^2 estimate; full-year training for the deployed model.

- Commercial: identical CBP/ZBP employment methodology as the original Coast `commercial.ipynb` (NAICS → ComStock-archetype mapping, CBECS-derived sqft/employee factors), reusing the already-statewide ZBP raw file (all Texas ZIP prefixes) rather than re-downloading it per zone.

9.4 Honest Accuracy Results

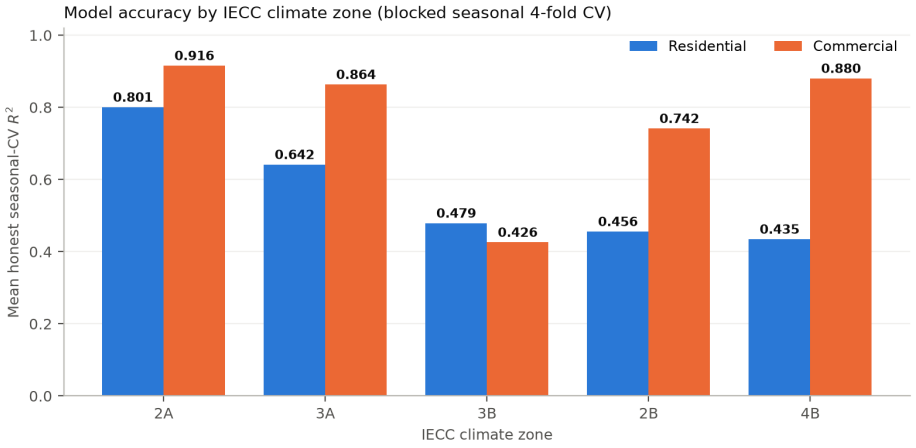


Figure 10: Mean honest seasonal-CV R^2 by IECC zone, residential vs. commercial.

Zone	Residential mean R^2	Commercial mean R^2
2A	0.801	0.916
3A	0.642	0.864
3B	0.479	0.426
2B	0.456	0.742
4B	0.435	0.880

Table 12: Honest seasonal-CV mean R^2, ranked by residential difficulty. Zone 2A (Coast's own climate) is clearly easiest; drier/more continental zones are harder, plausibly a mix of a more complex weather-load relationship (real heating seasons, not purely cooling-dominated) and smaller sample sizes. Values are from the current run; they shifted by up to 0.014 from the first statewide pass because the area-split allocation (Section 13) slightly changes each zone's county weather weights. Model choice, features and protocol are unchanged.

Commercial accuracy is generally higher than residential per zone, *except* in 3B, where it collapses to 0.426. Four zone/archetype combinations have a genuinely negative mean R^2: 3B `largehotel` (-1.36, as low as -3.14 in its worst season), 3B `smallhotel` (-0.28), 2B `smallhotel` (-0.26), and 3B `grocery` (-0.10). This was checked against a ComStock sample-size artifact and ruled out: 3B's `largehotel` archetype has *more* underlying sampled buildings (`models_used` = 881) than 2A's (361). The most likely explanation is that hotel demand in this region (Permian Basin) is driven by non-weather factors (oilfield workforce cycles) that a weather-only feature set cannot capture — a genuine "no signal" finding, not an artifact.

9.5 Cross-Check Against the Already-Validated Coast Pipeline

Because the new statewide grid does not share bus IDs with the original 200-bus Coast grid (Section 9), a bus-level comparison is not possible. Instead, the new zone-2A reconstruction is restricted to buses inside the original 12 Coast counties (1,505 matching buses on the full grid) and compared, at the aggregate level, against the already-validated Coast-only reconstruction (Section 6's `_featureeng` output).

Stack	Version	Peak (MW)	Annual mean (MW)	r
Residential	New (Coast-county subset, 1,505 buses)	11,552.6	4,895.1	0.9965
	Old (already-validated Coast pipeline)	10,084.0	4,260.3	
Commercial	New (Coast-county subset, 1,505 buses)	2,606.3	1,291.9	0.9993
	Old (already-validated Coast pipeline)	2,534.7	1,245.8	

Table 13: Cross-check: the new statewide methodology, restricted to the original 12 Coast counties, reproduces the already-validated Coast-only reconstruction closely on both stacks — the shape correlation is what this test is for, and it is essentially perfect. The level gap is larger on residential (+14.6%) than commercial (+2.8%) because the two footprints are not the same set: the full grid places 1,505 buses in those 12 counties against the old grid's 200, so the subset captures more of the counties' housing. The zone-wide weather blend (98 counties for the 2A run) diluting the Coast-specific 12-county blend also contributes.

9.6 Statewide Totals (Uncalibrated)

Zone	Residential peak (MW)	Commercial peak (MW)
2A	25,098.4	5,305.7
3A	17,101.3	3,343.4
3B	2,518.3	595.0
2B	1,020.1	294.1
4B	594.2	100.5
Statewide total (coincident)	**43,161.1**	**9,434.3**

Table 14: Per-zone and statewide peak load, residential and commercial, uncalibrated (no RES_SCALE/COM_SCALE applied) and with no industrial component anywhere except Coast's flat placeholder (not included in this table). The statewide row is the peak of the summed statewide series, not the sum of the per-zone peaks, which is why it falls below the column total: the five zones do not peak in the same hour.

10 Coast Recalibration

This section resolves the first open item from Section 16 of the previous phase.

10.1 Bug Discovery

Two independent problems were found while revisiting the Coast calibration (`plot_all.ipynb`). First, the grid-search cell contained a broken line (`results_matrix = .astype(float)`, a syntax error with the variable name missing) that could never have executed as written — meaning the notebook's stored outputs (Pearson 0.8806, RES_SCALE=1.9/COM_SCALE=1.3) were stale artifacts from some earlier working state, not reproducible from the code as it sat in the repository.

Second, and more importantly, every cell in this notebook was still reading the *original*, pre-fix profile files (no suffix), not `_featureeng` — meaning the "official" Coast calibration had never been re-run against any of the three corrections in Sections 3–6, exactly the staleness this open item had already flagged.

10.2 Fix and Results

The syntax error was fixed, every cell was switched to read `_featureeng`, and the grid search was re-run.

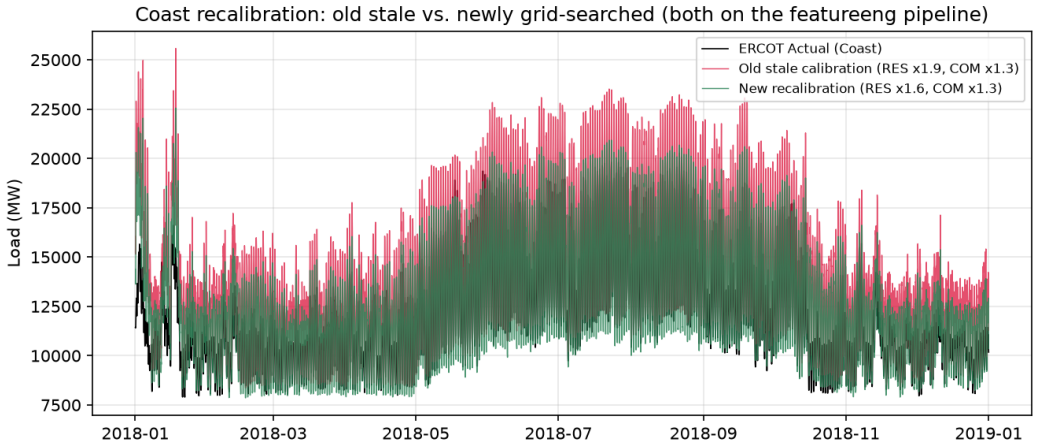


Figure 11: Old stale calibration vs. newly recalibrated total, both applied to the current best (`_featureeng`) pipeline, vs. ERCOT Coast actual.

Calibration	Peak (MW)	Pearson	MAE (MW)	RMSE (MW)
Old: RES 1.9 / COM 1.3 (stale)	25,577.8	0.9022	2,116.7	2,625.4
New: RES 1.6 / COM 1.3	**22,552.6**	**0.9024**	**1,201.4**	**1,550.9**

Table 15: ERCOT Coast actual peak: 20,269.8 MW. Pearson correlation is near scale-invariant and barely moves, but the old calibration was overcorrecting for a weaker, pre-fix pipeline: the absolute error nearly halves under the new calibration. **RES_SCALE=1.6, COM_SCALE=1.3 is the number to cite going forward.**

11 Validation Beyond Coast: an ERCOT Weather-Zone Crosswalk

Superseded by Section 14, which validates all eight ERCOT weather zones from a per-bus crosswalk contained in the grid model itself. This section is retained because its method — and the Panhandle finding below — document how the problem was approached before that field was available, and because its conclusion about West Texas was correct and is reproduced by the better method.

Coast is the only IECC zone with a direct, real validation number, because ERCOT publishes actual historical load by its own eight *weather zones* (`COAST`, `EAST`, `FWEST`, `NORTH`, `NCENT`, `SOUTH`, `SCENT`, `WEST`) — a completely different zoning scheme from NREL's five IECC *climate* zones used throughout this report. No public, machine-readable county-to-ERCOT-weather-zone crosswalk exists (confirmed via search: ERCOT publishes only a picture map, `ercot.com/news/mediakit/maps`). One

was built here by visually reading that map together with ERCOT's separate county-participation map (which does precisely show an in-ERCOT/not-in-ERCOT flag per county), anchored on the known latitude/longitude of the twelve cities labeled on the weather-zone map.

A notable finding while doing this: the Texas Panhandle is *not* uniformly outside ERCOT, as had been assumed earlier in this project. Roughly half its counties (including Amarillo/Potter/Randall) show as "in ERCOT" on the official county-participation map, which contradicts the common description of Amarillo as Southwest Power Pool (SPP) territory. Given this contradiction, IECC zone 4B was left out of this validation pass entirely rather than risk an overconfident partial claim. Extending precise ERCOT-zone validation to zones 2A (beyond Coast), 2B, 3A, and 4B appeared at the time to need real GIS shapefiles for ERCOT's weather zones, not a hand-read map — each of those zones straddles multiple ERCOT weather zones along boundaries that were not reliably resolvable visually. That turned out not to be the binding constraint: the grid model itself carries the assignment (Section 14).

11.1 Zone 3B: West Texas / Permian Basin

Zone 3B was chosen as the one unambiguous case: 67 of its 70 counties map cleanly to ERCOT's `WEST` and `FWEST` zones combined. The three excluded counties are El Paso and Hudspeth (genuinely outside ERCOT, in the Western Interconnection) and Hemphill (the one ambiguous Panhandle-adjacent corner). The zone-3B bus-level reconstruction (residential + commercial, Section 9) is restricted to these 67 counties' buses and compared to ERCOT's real `WEST+FWEST` total.

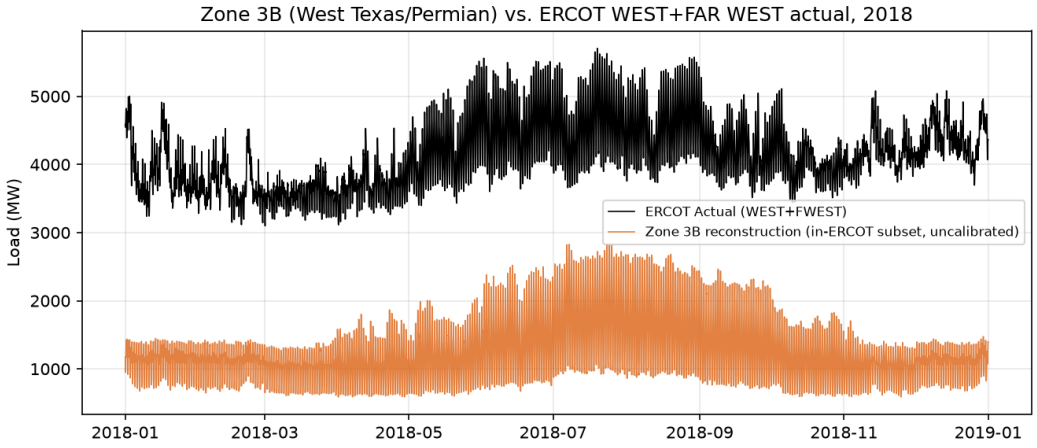


Figure 12: Zone 3B reconstruction (in-ERCOT county subset, uncalibrated) vs. ERCOT WEST+FAR WEST actual load, 2018.

	Zone 3B (vs. ERCOT WEST+FWEST)
ERCOT actual peak	5,704.2 MW
Reconstruction peak (uncalibrated)	2,827.2 MW
Uncalibrated peak coverage	49.6%
Pearson correlation	**0.71**

Table 16: Zone 3B vs. Coast for reference: Coast's uncalibrated peak coverage is ∼57% and its Pearson correlation is ∼0.90. The ∼57% figure is for the original 200-bus Coast-only pipeline; the same quantity measured on the statewide 6,717-bus grid is 65.3% (Table 20), the difference coming from the area-split allocation and the larger bus set, not from a change in the load models.

Magnitude coverage is in the same ballpark as Coast's own uncalibrated shortfall — no gross scaling bug — but the shape correlation (0.71) is meaningfully weaker than Coast's (∼0.90). The leading hypothesis is that West Texas load carries a much larger and more volatile *industrial* share (oilfield pumping and compression, driven by drilling activity rather than weather) than Coast's comparatively steady industrial baseload, and this reconstruction has no industrial model at all for 3B — not even Coast's flat placeholder. A weaker weather-only fit here is consistent with a genuinely missing, non-weather-driven demand driver, not a defect in the residential/commercial modeling itself (which, per Section 9, is honestly the weakest of the five zones already, independent of this comparison).

12 A Global, Statewide Validation Result

Independent of the county-level ERCOT-zone crosswalk (Section 11), a coarser but zero-assumption comparison is available: sum the residential and commercial reconstruction across *all five* IECC zones and compare directly to ERCOT's own published statewide total (the `ERCOT` column of the same native load file used throughout this report). This requires no crosswalk at all, since it is an aggregate-to-aggregate comparison — only a minor, quantifiable mismatch remains between the two footprints (the reconstruction includes some non-ERCOT counties, e.g. El Paso and parts of the Panhandle, that the ERCOT total excludes).

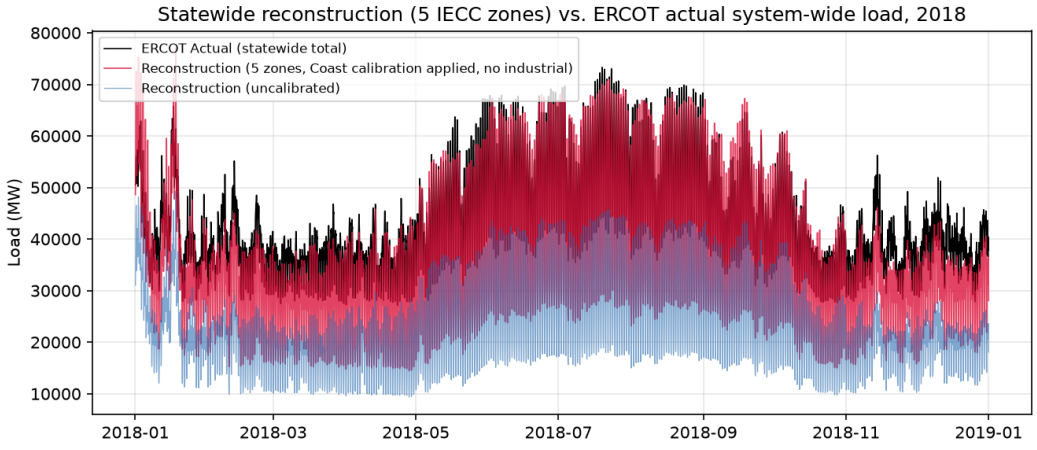


Figure 13: Statewide reconstruction (five IECC zones, residential + commercial) vs. ERCOT's actual statewide total load, 2018.

	Peak (MW)	Pearson
ERCOT actual, statewide total	73,308.2	—
Reconstruction, uncalibrated, no industrial	49,197.6 (67.1%)	0.9223
Reconstruction, Coast calibration applied (speculative)	76,856.9 (+4.8%)	0.9231

Table 17: The statewide aggregate correlates *more strongly* with ERCOT's true system total than any single zone does individually — including Coast.

This is a genuinely new result, not merely a rescaled version of an existing one: the statewide correlation (0.922) exceeds both Coast's own (0.90, Section 10) and zone 3B's (0.71, Section 11). The most plausible explanation is a diversity/smoothing effect — summing five climatically distinct zones averages out each zone's idiosyncratic noise and peak-timing offsets, directly analogous to the coincidence/diversity-factor concept already used for review point 5 (Section 7), just observed here at the statewide scale instead of within one zone. Applying Coast's calibration factors statewide, despite having no independent evidence they generalize (and zone 3B's much weaker local fit suggests they may not, zone-by-zone), still lands within 5% of ERCOT's true statewide peak — a reassuring, if speculative, consistency check. This result should be reported as directional: there is still no independently-validated statewide calibration, and no industrial model anywhere outside Coast's flat placeholder.

Both gaps are closed in Section 15, which fits a calibration and a data-driven industrial constant per ERCOT zone. Note also that the last row above is more serious than it first appears: Coast's factors reach 105% of the true peak with no industrial load included at all*, which is what motivated abandoning them as a statewide calibration.*

13 Spatial Allocation Refinement: From Nearest-Bus to Area-Split

Every result up to this point rested on a winner-takes-all spatial join: each census tract gave *all* of its dwellings to the single nearest bus, and each ZCTA gave *all* of its commercial floor space to the single nearest bus. This is the spatial-assignment limitation the paper lists as future work, and it is the dominant source of bus-level error — not the load models, which by then were already validated. Two new GIS exports replace it.

13.1 Method

Each 2020 census tract is subdivided into **50 equal-area parts**, and each 2010 ZCTA into **100 equal-area parts**; every part is then spatially joined to its own nearest bus. The allocation weight follows directly:

$$w(\text{polygon}, \text{bus}) \;=\; \frac{\text{parts won by that bus}}{\text{total parts}},$$

in 2% increments for tracts and 1% for ZCTAs. A polygon's dwellings and floor space are then distributed as a weighted sum over every bus that serves part of it, rather than dumped on one. The subdivision was verified rather than assumed: within a tract the 50 `Shape_Area` values agree to 10^{-6} and sum to `ALAND`; within a ZCTA the 100 values agree to 10^{-7}.

A third export, `texas7k_bus_with_climate_zone.csv`, supplies the authoritative bus table: all **6,717** Texas 7k buses with substation coordinates, nominal kV, the case's own `Load_MW`/`Gen_MW`, county FIPS, IECC zone, and the ERCOT `Area_Name`. Previously a bus existed in the pipeline

only if some tract had chosen it (3,426 did), and its climate zone had to be inferred by majority vote over those tracts. Two independent checks fall out of it: the file's own IECC zone agrees with the NREL-derived county crosswalk on **100% of 6,717 buses**, and `Load_MW` provides, for the first time, a per-bus reference against which a *spatial* allocation can be scored — something no aggregate ERCOT comparison can do.

13.2 Results

Weights sum to one per polygon by construction, so the allocation is conservative: **100.00% of Texas housing units** (11,628,419) are retained, none double-counted. Coverage and spatial agreement both improve substantially.

	Nearest-bus join	Area-split
Residential (50 parts per tract)		
Tracts split across >1 bus	0%	84.0%
Buses receiving dwellings	3,426	**6,352**
Load-carrying buses with zero dwellings	1,824	**25**
Spearman ρ vs. bus `Load_MW`	0.502	**0.606**
Commercial (100 parts per ZCTA)		
ZCTAs split across >1 bus	0%	90.2%
Buses receiving floor space	1,598	**6,329**
Load-carrying buses with commercial load	20.1%	**96.1%**
Spearman ρ vs. bus `Load_MW`	0.133	**0.475**

Table 18: Effect of the area-split allocation, scored against the Texas 7k case's own per-bus `Load_MW`. Spearman (rank) rather than Pearson is the meaningful statistic here: `Load_MW` also contains industrial demand, so a handful of very large industrial buses dominate any linear correlation.

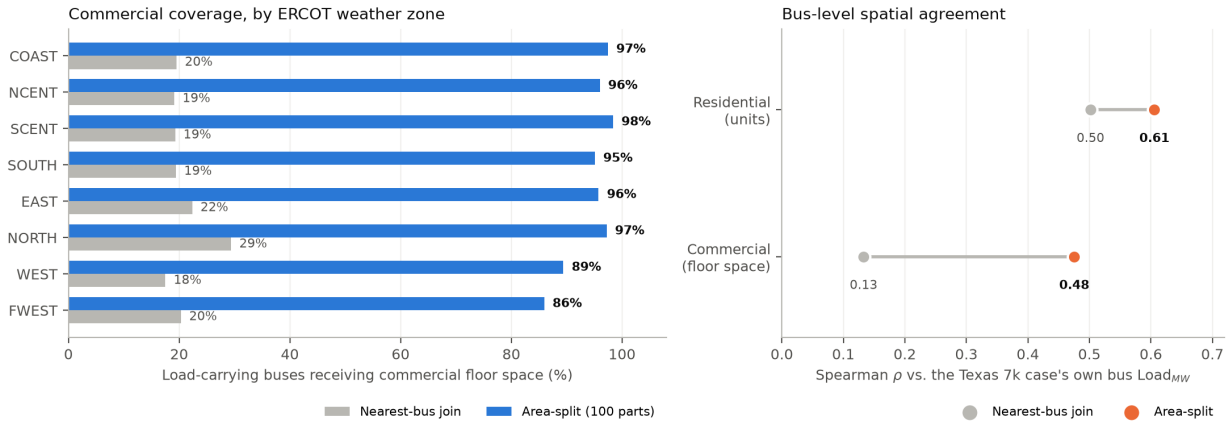


Figure 14: Left: share of load-carrying buses receiving commercial floor space, by ERCOT weather zone. Right: bus-level rank agreement with the case's own `Load_MW`.

The commercial gain is the larger one, because the nearest-ZCTA join was the more degenerate of the two: with only 2,031 ZCTAs to distribute over 6,717 buses, just 1,598 buses (23.8%) could

ever receive commercial load at all, in any zone — and only 20.1% of the buses that actually carry load in the grid case.

13.3 A Negative Result Worth Recording

The area split was expected to also repair a degenerate per-zone commercial calibration (Section 15, where South Central fits COM_SCALE = 0). **It did not**, and the reason is structural rather than empirical: redistributing floor space *among buses of the same ERCOT zone* leaves that zone's aggregate unchanged (zone totals moved by −1.6% to +7.8%, most under 2%), and the calibration is fitted on the aggregate. Every zone-level validation number in Sections 14 and 15 is, to three decimals, what it was before the ZCTA export landed.

This is worth stating plainly in the paper: the area split is a *bus-level* improvement, and bus-level fidelity is what the framework exists to produce, but it is invisible to any zone-aggregate metric. It is precisely the kind of improvement that a validation protocol based only on aggregate correlation would fail to detect — which is itself an argument for reporting the `Load_MW` rank statistic above.

14 ERCOT Weather-Zone Validation: All Eight Zones

Section 11 validated a single IECC zone (3B) against a county-to-ERCOT-weather-zone crosswalk read by eye from ERCOT's published picture map, and stated that extending this to the remaining zones would require GIS shapefiles that are not publicly available. **That obstacle is now removed**, from an unexpected direction: the authoritative bus table carries an `Area_Name` field holding exactly ERCOT's eight weather-zone names. The crosswalk is therefore per bus, taken from the grid model itself, with no county approximation and no map reading. The 3B result is superseded by what follows and is retained above only as methodology history.

14.1 Validating the Crosswalk Before Using It

`Area_Name` was checked before being trusted. The Texas 7k case is a peak-load snapshot, so each area's summed `Load_MW` should sit near that ERCOT zone's 2018 peak. Six of eight do; two clearly do not.

ERCOT zone	Case peak (MW)	ERCOT 2018 peak (MW)	Ratio
COAST	20,130.7	20,269.8	0.99
FWEST	3,558.9	3,655.0	0.97
NCENT	24,706.5	26,499.2	0.93
SOUTH	5,646.1	6,176.1	0.91
WEST	1,886.3	2,083.6	0.91
SCENT	11,899.7	14,166.7	0.84
EAST	3,473.7	2,620.8	**1.33**
NORTH	3,364.7	1,521.8	**2.21**
Statewide	74,666.6	73,308.2	1.02

Table 19: The synthetic case's EAST and NORTH area boundaries do not coincide with ERCOT's. Those two zones are reported throughout with a boundary-mismatch flag: their shape statistics remain interpretable, their magnitude statistics do not.

14.2 Results

The residential and commercial bus-level reconstructions from all five IECC zones are concatenated and grouped by each bus's ERCOT zone — an aggregation the previous county-based approach could not express, since a single ERCOT zone such as SCENT draws buses from IECC 2A and 3A at once.

ERCOT zone	ERCOT peak (MW)	Recon. peak (MW)	Peak coverage	Pearson r
COAST	20,269.8	13,233.6	65.3%	0.900
NCENT	26,499.2	17,000.0	64.2%	0.900
SCENT	14,166.7	9,039.0	63.8%	0.887
SOUTH	6,176.1	3,993.7	64.7%	0.909
WEST	2,083.6	914.5	43.9%	0.815
FWEST	3,655.0	2,299.7	62.9%	**0.629**
EAST†	2,620.8	3,494.2	—	0.911
NORTH†	1,521.8	1,730.6	—	0.905
ERCOT total	73,308.2	49,197.6	67.1%	**0.922**

Table 20: Uncalibrated reconstruction (no scale factors, no industrial model) against ERCOT actual load, 2018. † boundary mismatch: peak coverage is not meaningful for these two zones.

Three findings. First, the framework holds up across climatically distinct regions: four of the six well-matched zones (COAST, NCENT, SCENT, SOUTH) land between $r = 0.887$ and $r = 0.909$. Uncalibrated peak coverage is more uniform still — five of the six sit between 62.9% and 65.3%, WEST being the sole exception at 43.9% — and that consistency suggests one systematic missing component rather than per-zone modeling noise.

Second, the statewide aggregate ($r = 0.922$) again exceeds every individual zone, the diversity/smoothing effect already reported in Section 12, now measured against a correctly-partitioned set of zones rather than an approximate one.

Third, **FWEST is the clear outlier** ($r = 0.629$), with WEST second ($r = 0.815$). This confirms, far more precisely than the earlier county-approximated 3B result ($r = 0.71$ over a hand-built 67-county subset), that the weakness is specific to West Texas and the Permian Basin rather than a general property of drier climate zones.

15 Per-Zone Calibration with a Fitted Industrial Constant

Coast's recalibrated factors (Section 10: RES_SCALE = 1.6, COM_SCALE = 1.3, plus a hand-picked flat 4,500 MW industrial term) were fitted on Coast alone. A direct test shows they do not transfer: applied statewide *with no industrial term at all*, they already reach 76,857 MW, or **105% of ERCOT's true statewide peak**. The residential scale factor had evidently been absorbing missing industrial demand.

15.1 Method

Each ERCOT zone is calibrated independently under

$$\mathrm{ERCOT}_z(t) \;=\; \mathrm{RES}_z \cdot r_z(t) \;+\; \mathrm{COM}_z \cdot c_z(t) \;+\; \mathrm{IND}_z,$$

solved by non-negative least squares. Two departures from the Coast method matter. The industrial term is **fitted rather than assumed**, making IND_z a data-driven estimate of each zone's weather-

insensitive baseload; and the fit is scored by **blocked seasonal 4-fold cross-validation** (fit on three seasons, score on the held-out one), the same protocol used for the load models throughout this report. A calibration fitted and scored on the same 8,760 hours would be optimistic in precisely the way diagnosed in Section 3.

Collinearity was checked before interpreting any individual coefficient: on centered-and-scaled columns the design matrix has condition number ≈ 2.3 and VIF ≈ 1.9, so the residential/commercial split is well posed.

15.2 Results

ERCOT zone	RES	COM	IND (MW)	CV MAE (MW)	Pearson r
COAST	1.07	0.98	5,594	1,105 (from 1,933)	0.900
NCENT	1.36	0.98	4,863	1,496 (from 1,884)	0.901
SCENT	1.26	**0.00**	2,531	715 (from 1,002)	0.902
SOUTH	1.14	0.66	1,541	304 (from 578)	0.911
WEST	1.07	**3.03**	692	124 (from 227)	0.823
FWEST	**0.30**	1.46	2,236	213 (from 602)	0.655
EAST†	0.58	0.70	642	122 (from 223)	0.911
NORTH†	0.58	0.77	393	69 (from 125)	0.905
ERCOT total	**1.15**	**0.76**	**18,079**	**3,462** (from 6,085)	**0.924**

Table 21: Per-zone calibration. "from" gives the blocked-seasonal-CV MAE of the Coast-style two-parameter fit with no industrial term, for comparison. † boundary mismatch, reported for completeness.

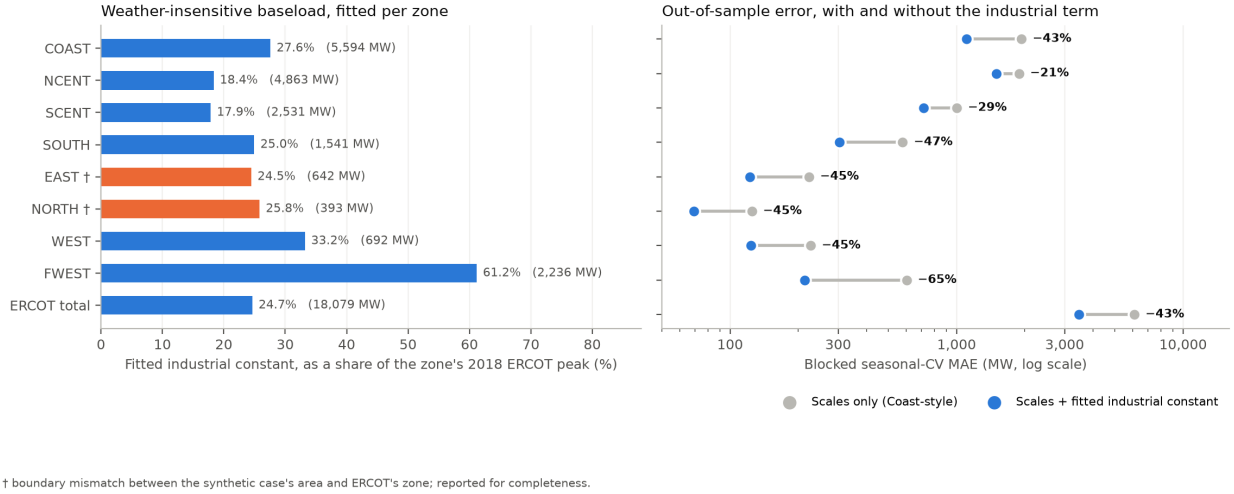


Figure 15: Left: the fitted weather-insensitive baseload as a share of each zone's ERCOT peak. Right: out-of-sample error with and without the industrial term.

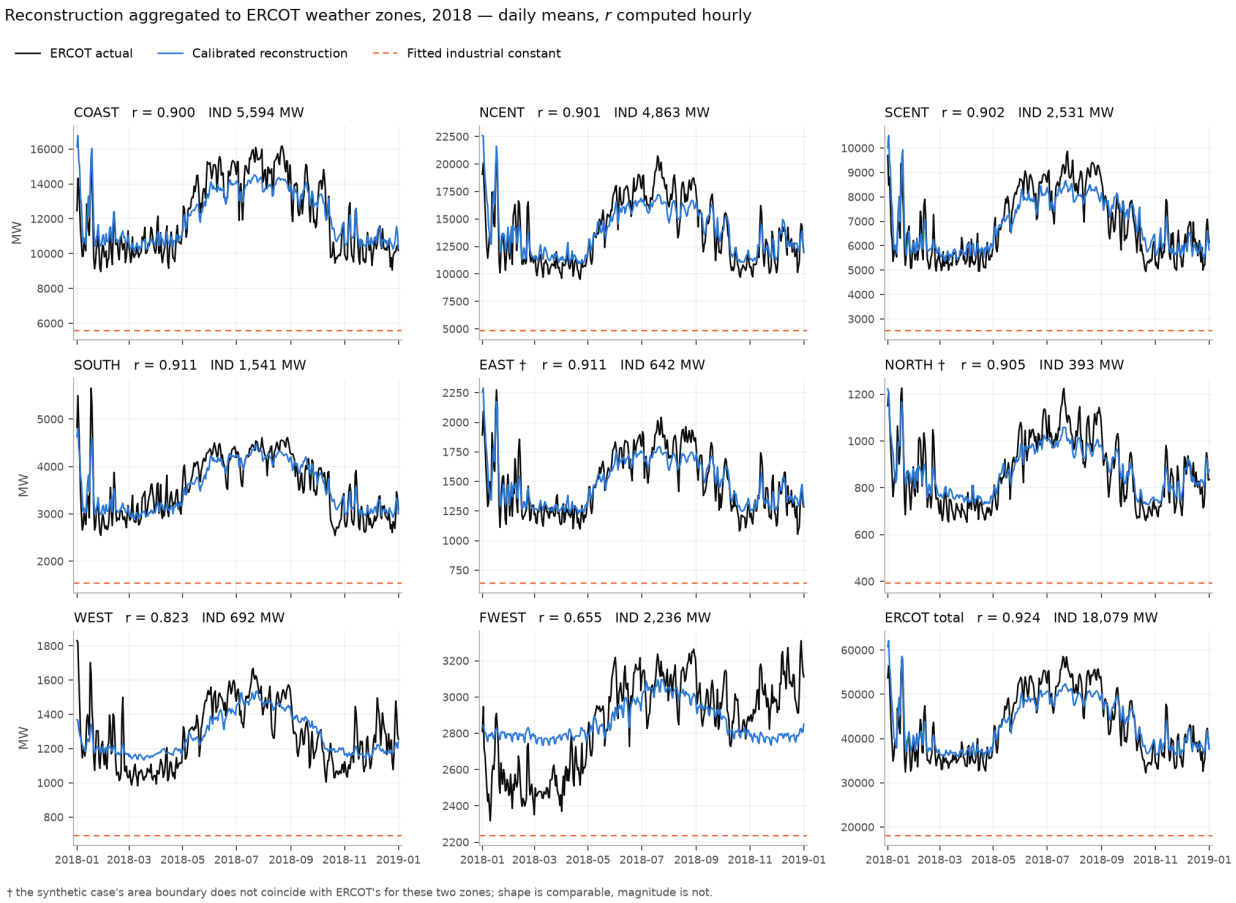


Figure 16: Calibrated bus-level reconstruction aggregated to each ERCOT weather zone, 2018. Daily means are plotted for legibility; the reported r is computed on hourly data.

Four results stand out.

The building-stock reconstruction barely needs a fudge factor. Once an industrial term is present, RES falls to 1.07–1.36 in five of the six well-matched zones (FWEST, at 0.30, is the exception discussed below), against 1.6–1.9 for Coast without one. This is a meaningful argument for the method: the earlier large scale factors were compensating for a missing load component, not for a mis-sized housing stock.

Industrial demand is roughly a quarter of ERCOT. The statewide fit puts it at 18,079 MW, or 24.7% of the system peak. Coast alone fits 5,594 MW, which retrospectively validates the paper's hand-picked 4,500 MW placeholder as the right order of magnitude and slightly low.

FWEST is structurally industrial. It fits RES = 0.30 and an industrial constant equal to **61.2% of its peak** — by far the highest share of any zone, and a quantitative explanation for its weak correlation. Its load is mostly not a function of weather, so no weather-driven reconstruction can track it.

Out-of-sample error drops sharply, by 21% (NCENT) to 65% (FWEST) per zone, and by 43% statewide (blocked-seasonal-CV MAE 6,085 → 3,462 MW).

15.3 Two Caveats, One Unresolved

A constant added to an *already-fixed* prediction cannot change a correlation, since $\text{corr}(y, ax + b)$ does not depend on b. Here IND is fitted *jointly* with the scales, which does move the shape slightly — without an industrial term the scales are forced upward to reach the right level, distorting the residential/commercial mix. The effect is real but small (FWEST 0.631 → 0.655, WEST 0.807 → 0.823, every other zone +0.002). The industrial constant should therefore be reported as a magnitude correction, not as a fix for the weak-shape zones.

Two coefficients remain anomalous: SCENT fits COM_SCALE = 0.00 and WEST fits 3.03. The obvious explanation — poor commercial bus coverage — was tested and rejected (Section 13). The real diagnosis, for SCENT, is that the *partial* correlation of the commercial profile with ERCOT load, with residential removed, is −0.014, against +0.24 to +0.34 in every other zone: the commercial stack genuinely adds no information there beyond residential plus a constant. Why this should be true of the Austin/San Antonio corridor specifically is **not yet explained**, and is flagged as an open item rather than resolved.

16 Open Items and Next Steps

- **RES_SCALE/COM_SCALE recalibration — DONE (Section 10).** New calibration RES_SCALE=1.6, COM_SCALE=1.3, roughly halving the absolute error (MAE 2,117 MW → 1,201 MW) against the current best pipeline.
- **Extension to all of Texas — DONE (Section 9).** The missing bus-to-load spatial mapping has since been obtained for both residential and commercial, and subsequently replaced by the full 6,717-bus authoritative table with an area-split allocation (Section 13). All five IECC zones now have honest R^2 results and are cross-checked against Coast at $r > 0.996$.
- **Weekday/weekend breakdown of residential R^2**: Section 7 shows the winter-weekend diurnal-shape correlation is the single weakest metric in this report, and that it was fixed by the weather correction specifically, not by later refinements. A direct weekday/weekend breakdown of the seasonal-CV R^2 itself (rather than only the aggregate shape correlation) has not yet been produced. Still not done; now also an open question for the four other statewide zones.
- **Comparison against prior proportional allocation methods** (review point 9, "Hanyue's method"): still blocked on obtaining the method reference itself.
- **Spatial diversity between buses**: all buses within a zone still share a single, zone-wide blended weather series (Section 4). Still not implemented; the statewide extension makes this a bigger question, since a 79-county zone-wide blend (2A) dilutes local signal more than Coast's original 12-county blend did (quantified in the ∼5% cross-check gap, Table 13).
- **Commercial bus-coverage gap — DONE (Section 13).** The nearest-ZCTA join left four buses in five unable to receive any commercial load. An area-split join (100 equal-area parts per ZCTA, each matched to its nearest bus) raises coverage of load-carrying buses from 20.1% to 96.1% and rank agreement with the case's own `Load_MW` from $\rho = 0.133$ to 0.475. The equivalent split for tracts (50 parts) cut load-carrying buses with zero dwellings from 1,824 to 25.

- **Industrial load, statewide — PARTIALLY ADDRESSED (Section 15).** Each ERCOT zone now carries a *fitted* weather-insensitive constant instead of a hand-picked one, totalling 18,079 MW statewide (24.7% of system peak), which roughly halves out-of-sample error. But a constant is still not a model: it cannot reproduce industrial load's own temporal variation, which is precisely what FWEST needs ($r = 0.629$, with a fitted constant equal to 61.2% of its peak). A genuinely time-varying, data-driven industrial representation remains the single most consequential open item.
- **ERCOT weather-zone crosswalk — DONE (Section 14).** No shapefiles were needed in the end: the grid model's own `Area_Name` field assigns every bus to an ERCOT weather zone. All eight zones plus the ERCOT total now have a real validation number.
- **EAST and NORTH boundary mismatch (NEW, Section 14):** the synthetic case's area load for these two zones is 1.33× and 2.21× their ERCOT peak, so the case's boundaries there do not coincide with ERCOT's. Their shape statistics are reported, their magnitudes are not. Diagnosing which counties are misassigned would make two more zones fully usable; not attempted.
- **South Central's null commercial coefficient (NEW, Section 15):** SCENT fits COM_SCALE = 0, and the partial correlation of its commercial profile with ERCOT load (residential removed) is −0.014, against +0.24 to +0.34 everywhere else. Coverage and collinearity were both tested and rejected as explanations. Unexplained; isolated to the Austin/San Antonio corridor.
- **Model choice for the paper**: Section 5 supports either retaining RandomForest (simpler, statistically equivalent system-level result) or adopting LightGBM (materially better per-archetype accuracy at zero system-level cost). The statewide extension (Section 9) used LightGBM throughout, so this is effectively the working default now, but the framing decision for the paper text itself is still open.

16.1 What Would Help This Move Faster

- **Real ERCOT weather-zone GIS shapefiles — no longer needed.** Previously listed here as the single highest-leverage missing input; the grid model's `Area_Name` field supplied the assignment instead (Section 14). Retained here only to record that the ask is closed.
- **A data source and methodology for industrial load**, statewide — now with a target to hit. Section 15 quantifies how much weather-insensitive load each zone is missing (18,079 MW statewide; 61.2% of peak in FWEST against 17.9% in SCENT), which confirms that industrial share varies enormously by region and gives any candidate industrial dataset a per-zone number to be checked against.
- **The Hanyue proportional-allocation method reference** (review point 9), to unblock the one remaining unimplemented item from the point-5 validation work.
- **A decision on scope for the paper**: whether the statewide extension (Sections 9–12) belongs in the current submission as an additional contribution, or is better positioned as follow-up work referenced briefly — this materially changes how much of the above needs to be resolved before submission versus after.

17 Conclusion

Two independent, previously undiagnosed issues in the original pipeline — temporal train/test leakage and an incorrect weather station geography — were identified, root- caused, and corrected under a single, consistent, honest evaluation protocol (blocked seasonal 4-fold cross-validation). Both corrections improved, rather than hurt, the paper's headline system-level validation metric (Pearson correlation against ERCOT Coast actual load), while substantially lowering the previously-inflated per-archetype R^2 figures to their honest values. A model comparison (RandomForest/XGBoost/LightGBM) and a feature-engineering ablation study, both run through the same protocol, directly address review points 5 and 6: model choice improves per-archetype accuracy but not the system-level number, while a cheap lag-temperature feature addition improves both. A final full comparison across all six pipeline versions (Section 7) shows this improvement is not perfectly uniform across every secondary metric — the temporal and weather fixes did most of the work on shape-sensitive metrics like the winter-weekend diurnal correlation, while model choice and feature engineering mainly improved accuracy and the headline correlation. The corrected, feature-engineered LightGBM pipeline is recommended as the version to report in the paper going forward.

This corrected pipeline was then extended from the single Coast case study to all five Texas IECC climate zones (Section 9), reusing the exact methodology validated on Coast, and cross-checked against it at $r > 0.996$ on both the residential and commercial stacks. The Coast calibration itself was found to be stale and was refreshed (Section 10), materially reducing absolute error.

Three further results complete the picture. First, the winner-takes-all spatial join that underpinned every earlier number — one tract, one bus — was replaced by an area-split allocation (Section 13), conserving 100% of Texas housing units while raising the share of load-carrying buses that receive commercial load from 20% to 96% and roughly quadrupling bus-level rank agreement with the grid case's own per-bus load. Notably, this improvement is invisible to every zone-aggregate metric in this report, which is itself an argument for reporting a bus-level statistic.

Second, validation was extended from one hand-crosswalked zone to *all eight* ERCOT weather zones (Section 14), using a per-bus assignment carried by the grid model itself rather than the unobtainable GIS shapefiles previously assumed necessary. Four of the six well-matched zones fall between $r = 0.887$ and $r = 0.909$, five of the six show uncalibrated peak coverage within 62.9–65.3%, and the statewide aggregate again leads at $r = 0.922$. The earlier zone-3B finding is reproduced and sharpened: the weakness is specific to West Texas ($r = 0.629$ for FWEST), not to dry climates generally.

Third, calibration was refitted per zone with a *data-driven* industrial constant (Section 15), scored out-of-sample by the same blocked seasonal protocol. Coast's factors were shown not to transfer — applied statewide with no industrial term they already reach 105% of ERCOT's true peak — while the refitted residential scale factors collapse to 1.07–1.36 in five of the six well-matched zones, indicating that the earlier large factors were standing in for missing industrial demand rather than correcting the housing stock. That missing demand is now quantified at 18,079 MW statewide, 24.7% of system peak, and out-of-sample error roughly halves.

Together, these results show the framework generalizes well beyond its original single-zone case study, while being explicit about where it does not yet hold: industrial load is a fitted constant rather than a model, and a constant cannot rescue the one zone that most needs it (FWEST, whose weather-insensitive share is 61% of peak); two ERCOT zones (EAST, NORTH) have boundary mismatches that make their magnitudes uninterpretable; and South Central's null commercial coefficient remains unexplained after coverage and collinearity were both tested and ruled out. Section 16.1 lists what would most efficiently close the remaining gaps.